

E Albert Einstein Cup 2024

Session 1: Implementation

Introduction

The designer has provided you with mockups (*Mockups.xd*). As a mobile app developer, your task is to develop the corresponding app in strict accordance with the design and requirements of these mockups. You have **3.5 hours** to implement the app for the provided phone.

General Demands

1. Strictly follow the details of the mockups, including different states of interactive elements.
2. The app should be called "Events NB" and use the provided icon (*Icon.png*) in the system launcher.
3. The app starts on the Event List Page with the Grid tab selected.
4. The bottom navigation bar should be shown on all pages. Tapping on the first icon navigates to the Event List Page (Grid tab), tapping on the second icon navigates to the Ticket List Page (Upcoming tab).



Page Demands

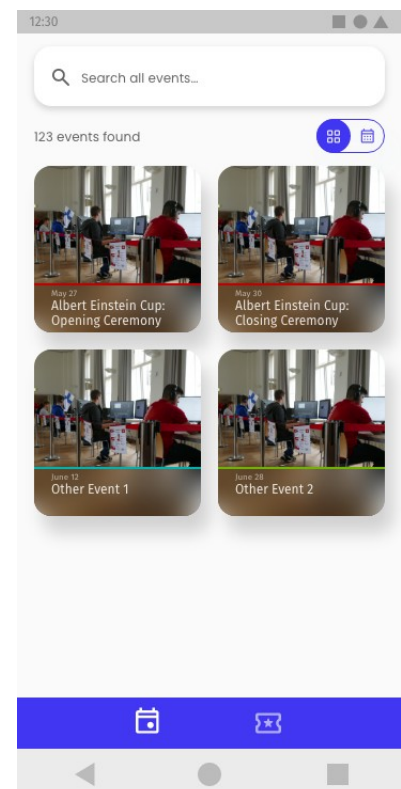
1 Event List Page

Elements

The Event List Page consists of two tabs: Grid and Calendar. On both tabs, there's a search bar at the top, a text showing how many events were found, and a tab switcher.

The Grid tab shows a vertically scrollable grid of upcoming events. Each event consists of a name, date, colour, and image.

The Calendar tab shows a horizontally swipeable calendar with all events. Each page of the calendar covers one month (and potentially a few days before or after) with rows for every week (starting on Monday). Each event consists of a name and colour.



Functional Requirements

1. The search does not have to be implemented.

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- The user can switch between both tabs using the icon buttons below the search bar.
- Both tabs should use the provided sample data (*Data.json*).
- Tapping on an event should open the Event Details Page for that event.
 - If the user taps on an event in the Grid tab, the event's image should animate to its new position on the Event Details Page.
- Grid Tab:
 - The event grid is vertically scrollable.
 - Only current and upcoming events are shown.
 - Events are ordered by their date (first one at the top).
- Calendar Tab:
 - The calendar is horizontally swipeable, Each page of the calendar covers one month (and potentially a few days before or after) with rows for every week (starting on Monday). Initially, the calendar starts with the current month.
 - All events are shown.

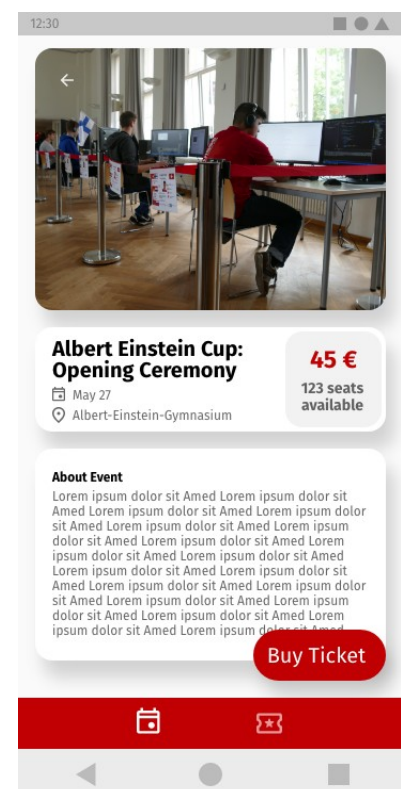
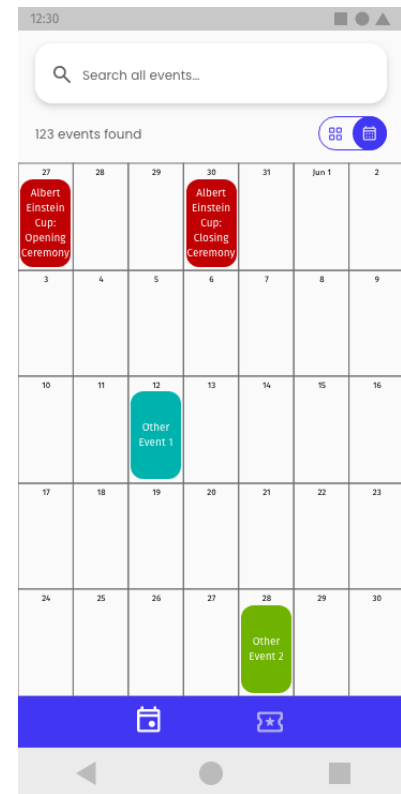
2 Event Details Page

Elements

The Event Details Page shows all details of an event and allows the user to start the ticket booking process. They can also go back to the previous page.

Functional Requirements

- The back button leads to the previous page.
- Show the event's image, name, date, location, price, available seat count, and description ("About Event").
- The price and Buy Ticket button, as well as the bottom navigation bar, use the event's colour.
- The Buy Ticket button navigates to the Buy Ticket Page for this event. Only show this button if the event is today or in the future.



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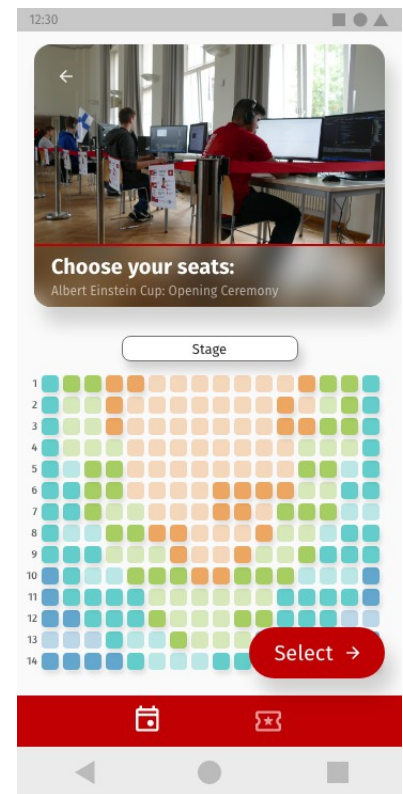
3 Buy Ticket Page: Choose Your Seats

Elements

The Buy Ticket Page: Choose Your Seats allows users to select their seats for an event from a grid with row numbers and all seats. They can also go back to the previous page.

Functional Requirements

1. The back button leads to the previous page.
2. Show the event's image and name.
3. Each seat in the seat grid has the correct class (A, B, C, or D, as indicated by its colour). Refer to the mockup for the correct assignment of seats to seat classes.
 1. Seats that are already booked have a lowered opacity and can't be selected.
 2. Seats can be selected and deselected by tapping on them. Selected seats have a different design.
4. The line on top of the event's image, the Select button, and the bottom navigation bar use the event's colour.
5. The user can continue to the Buy Ticket Page: Enter Your Details using the Select button. Only show this button if at least one seat was selected.



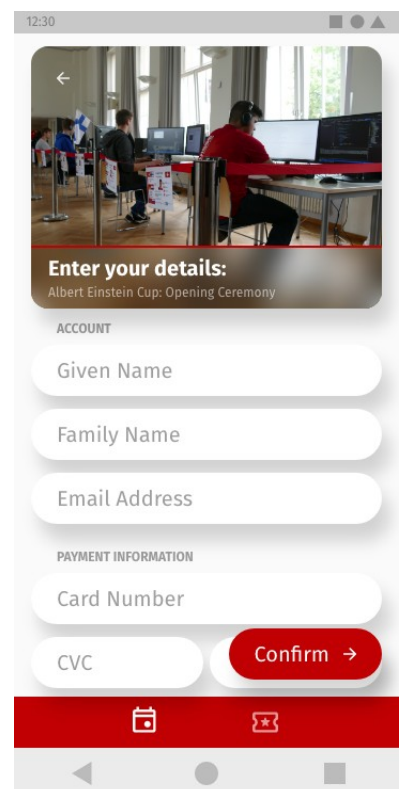
4 Buy Ticket Page: Enter Your Details

Elements

The Buy Ticket Page: Enter Your Details allows users to enter their account and payment information to complete the ticket booking. They can also go back to the previous page.

Functional Requirements

1. The back button leads to the previous page. The selected seats should be remembered.
2. Show the event's image and name.
3. All text fields are required (may not be empty) to complete ("Confirm") the booking.
 1. The card number must be exactly 16 digits.
 2. The CVC must be exactly 3 digits.



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3. Clicking on the Expiration field opens a dialog in which the user can select the expiration year and month of their credit card (must be in the future). The design is up to you and should be consistent with the rest of the mockups.
4. Show appropriate error messages for all fields.
4. The line on top of the event's image, the Confirm button, and the bottom navigation bar use the event's colour.
5. When they have entered all information, the user can click "Confirm". This navigates to the Ticket List Page (Upcoming tab), focused on their new ticket.

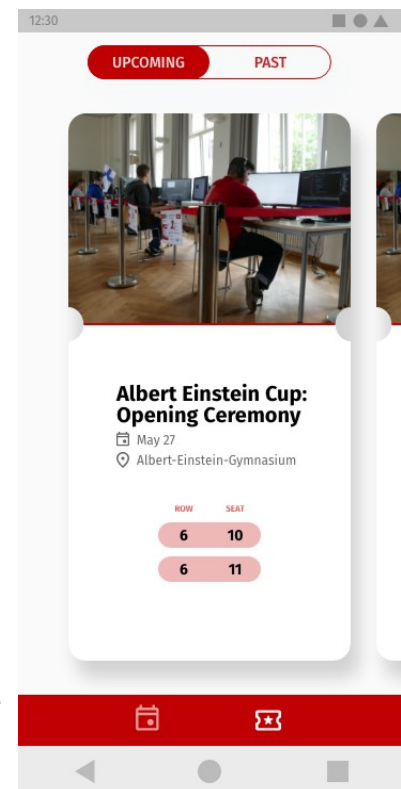
5 Ticket List Page

Elements

The Ticket List Page has two tabs that show the user's today's/upcoming tickets and past tickets, respectively.

Functional Requirements

1. The user can switch between the Upcoming and Past tabs using the tab indicator at the top.
2. The user can horizontally swipe through their tickets.
3. For each ticket, show the event's image, name, date, and location, as well as the booked seats.
4. The line between the event's image and the lower part of the ticket should use the event's colour. The row and seat info should use the event's colour, but using the opacities from the mockup.
5. The tab indicator and bottom navigation bar should use the currently visible event's colour. While swiping to a different event, animate the colour change.
6. For past events, the event's image should have a lower opacity (see the mockup).



Submission

Your submission should contain the following top-level files and directories:

- the compiled APK
- a directory with the full source code
- optional: a README file containing notes for the marking

These files and directories should then be pushed to the remote Git repository.